# DAT151: Assignment 6 Report

**Group:** 5

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## Task 1: LVM

In assignment 0 you were asked to keep at least 10GiB free space on the disk. You will need that free space in this task. If the free space was assigned to a partition during install, the installation program will have set up a file system on the partition. You must remove the file system before you can use the partition with LVM. If you did not leave 10GiB of free space on the disk, you must reinstall AlmaLinux on the computer, i.e. redo all of assignment 0. Actions to remove the file system of the partition are:

- If the partition is mounted, unmount the partition.
- If the partition is listed in the /etc/fstab file, comment out or remove the entry from that file.
- If you have created a systemd mount unit for the partition, delete the mount unit file and reload the daemons.
- Use the dd tool to write zeros (/dev/zero) to the first 100MiB of the partition, e.g. /dev/sda4.
- If you pick the wrong partition, you will destroy your system!
- In order to recover if you clear the wrong partition, first use the dd tool to dump the first 100MiB to a file, e.g. /root/origs/sda4.img.
- Use e.g. the tool parted and delete the partition. On the empty disk space, create a physical partition but leave 5GiB of empty space. The remaining 5GiB of empty space should for now not be assigned to a physical partition. Set up the physical partition as a LVM physical volume (PV). Create a new logical volume group (VG) using the new PV. When you have setup the VG, create some LVs, and make the LVs available to the system. Then save some files on the LVs. After setting up LVM, increase the volume group by 5 GiB. Describe and explain each step taken.

### a) Initial disk state and partition creation

We found our 10GiB Partition (sda4).

```
ffeilke@host67:~$ lsblk
NAME                          MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda                             8:0    0 232.9G  0 disk
├─sda1                          8:1    0   600M  0 part /boot/efi
├─sda2                          8:2    0     1G  0 part /boot
├─sda3                          8:3    0 220.8G  0 part
│ ├─almalinux_server67a-root 253:0    0   213G  0 lvm  /
│ └─almalinux_server67a-swap 253:1    0   7.8G  0 lvm  [SWAP]
└─sda4                          8:4    0  10.5G  0 part
sr0                            11:0    1  1024M  0 rom
```

```
ffeilke@host67:~$ sudo fdisk -l /dev/sda4
Disk /dev/sda4: 10.51 GiB, 11282677760 bytes, 22036480 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
ffeilke@host67:~$ sudo fdisk -l /dev/sda
[sudo] password for ffeilke:
Sorry, try again.
[sudo] password for ffeilke:
Disk /dev/sda: 232.89 GiB, 250059350016 bytes, 488397168 sectors
Disk model: Samsung SSD 850
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 1E645A9A-E132-474E-9F21-CD10D33EBD72

Device          Start       End   Sectors   Size Type
/dev/sda1        2048   1230847   1228800   600M EFI System
/dev/sda2     1230848   3327999   2097152     1G Linux extended boot
/dev/sda3     3328000 466360319 463032320 220.8G Linux LVM
/dev/sda4   466360320 488396799  22036480  10.5G Linux filesystem
```

We unmounted sda4:

```
unmount /dev/sda4
```

We also checked the /etc/fstab file:

```
sudo vim /etc/fstab
```

And checked for costum .mount files:

```
ls /etc/systemd/system/*.mount
```

To have a backup, we saved the first 100MiB before clearing the filesystem. We used the dd tool to write zeros (/dev/zero) to the first 100MiB of the Partition. In our next step we deleted the partition and returned the free space.

```
ffeilke@host67:~$ sudo mkdir -p /root/origs
ffeilke@host67:~$ sudo dd if=if/dev/sda4 of=/root/origs/sda4.img bs=1M count=100
dd: failed to open 'if/dev/sda4': No such file or directory
ffeilke@host67:~$ sudo dd if=/dev/sda4 of=/root/origs/sda4.img bs=1M count=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.198391 s, 529 MB/s
ffeilke@host67:~$ sudo dd if=/dev/zero of=/dev/sda4 bs=1M count=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.23204 s, 452 MB/s
ffeilke@host67:~$ sudo parted /dev/sda rm 4
Information: You may need to update /etc/fstab.
```

We createt a physical partition of 5GiB to leave 5GiB of empty space.

```
ffeilke@host67:~$ sudo fdisk /dev/sda

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.


Command (m for help): n
Partition number (4-128, default 4): 4
First sector (466360320-488397134, default 466360320):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (466360320-488397134, default 488396799): +5G

Created a new partition 4 of type 'Linux filesystem' and of size 5 GiB.

Command (m for help): w
The partition table has been altered.
Syncing disks.
```

## b) Create PV, VG and LVs

We set up the physical partition as a LVM physical volume (PV) and created a logical volume group (VG).

```
ffeilke@host67:~$ sudo pvcreate /dev/sda4
  Physical volume "/dev/sda4" successfully created.
ffeilke@host67:~$ sudo vgcreate assignment6_vg /dev/sda4
  Volume group "assignment6_vg" successfully created
```

We createt a logical volume. Formated it and mounted it.

```
ffeilke@host67:~$ sudo lvcreate -L 2G -n lv_data assignment6_vg
  Logical volume "lv_data" created.
ffeilke@host67:~$ sudo mkfs.ext4 /dev/assignment6_vg/lv_data
mke2fs 1.47.1 (20-May-2024)
Discarding device blocks: done
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: f72e857a-4b25-4cec-b446-6ee382d07e4c
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

ffeilke@host67:~$ sudo mkdir /mnt/assignment6
ffeilke@host67:~$ sudo mount /dev/assignment6_vg/lv_data /mnt/assignment6
ffeilke@host67:~$ echo "LVM is working" | sudo tee /mnt/assignment6/test.txt
LVM is working
```

### c) Extend VG with remaining 5GiB

We createt a new partition of the remaining free space and extended the VG with this new partition. In the end we varify that the volume group has been extendet.

```
ffeilke@host67:~$ sudo fdisk /dev/sda

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.


Command (m for help): n
Partition number (5-128, default 5):
First sector (476846080-488397134, default 476846080):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (476846080-488397134, default 488396799):

Created a new partition 5 of type 'Linux filesystem' and of size 5.5 GiB.

Command (m for help): w
The partition table has been altered.
Syncing disks.

ffeilke@host67:~$ sudo pvcreate /dev/sda5
  Physical volume "/dev/sda5" successfully created.
ffeilke@host67:~$ sudo vgextend assignment6_vg /dev/sda5
  Volume group "assignment6_vg" successfully extended
ffeilke@host67:~$ sudo vgs
  VG                 #PV #LV #SN Attr   VSize    VFree
  almalinux_server67a   1   2   0 wz--n- <220.79g    0
  assignment6_vg        2   1   0 wz--n-   10.50g 8.50g
```

## Task 2: NFS

For this task you will need two computes, one for the NFS server and the other as the NFS client. Make sure that the user account exists in the LDAP (task 2 of assignment 5), and that the user can log in using SSSD (task 3 of assignment 5). Move the user home directory to /share/home on the NFS server. The user home directory should not exist on the client. Set up the NFS server and export the directory /share/home to the client. The exported directory should be mounted as /share/home on the client. If necessary, modify the LDAP property homeDirectory to be consistent with a home directory below /share/home. Configure the NFS server to start at boot, and configure the client to mount the exported directory at boot. You can either create a systemd mount unit, or modify the /etc/fstab file. On the client, copy or make some changes to a file in the NFS mounted directory, and then check the attributes of that file on the server. Please document all you have done for configuring the servers and clients. Observe that you may need to adjust the firewall settings and/or configure SELinux to make NFS mount work. For SELinux, see e.g. Adjusting the policy for sharing NFS and CIFS volumes by using SELinux booleans. For your information, the system can be configured to mount the home directory of a user when the user logs in, and unmount on logout. This can be very useful on a system with many users, to save resources. Automatic mounting of home directories can be achieved e.g. through PAM by the pam_mount(3) module, or by an auto mounting system. Several auto mounting systems are available for Linux, e.g. using a systemd autmount unit, or the autofs daemon. Observe, you are not asked to configure your system to mount home directories on user login. It is sufficient in this task to mount all home directories at boot.

## a) Server configuration

We installed the necessary packages on both the server and the client. Next, we created the directory and moved the user home directory to /share/home.

```
sudo dnf install nfs-utils -y
```

```
soukup@host68:~$ sudo mkdir -p /share/home
```

```
soukup@host68:~$ sudo mv /home/testuser /share/home/
[sudo] password for soukup:
```

Opening `nfs`, `rpc-bind`, and `mountd` services in firewall:

```
soukup@host68:~$ sudo firewall-cmd --permanent --add-service=nfs
success
soukup@host68:~$ sudo firewall-cmd --permanent --add-service=rpc-bind
success
soukup@host68:~$ sudo firewall-cmd --permanent --add-service=mountd
success
```

Configure export rule for client IP in `/etc/exports`:

```
/share/home 10.0.0.67(rw,sync,no_root_squash,no_subtree_check)
~
```

We enable the `nfs-server` and run `exportfs -rav` so our changes will be considered.

```
soukup@host68:~$ sudo systemctl enable --now nfs-server rpcbind
Created symlink '/etc/systemd/system/multi-user.target.wants/rpcbind.service' → '/usr/lib/systemd/system/rpcbind.service'.
soukup@host68:~$ sudo exportfs-rav
sudo: exportfs-rav: command not found
soukup@host68:~$ sudo exportfs -rav
exporting 10.0.0.67:/share/home
```

## b) Modify LDAP User

We used an LDIF file to update the homeDirectory pointer in the LDAP database.

```
GNU nano 8.1                                                    modify_path.ldif
dn; uid=testuser,,ou=People,dc=h68,dc=dat151
changetype: modify
replace: homeDirectory
homeDirectory: /share/home/testuser
```

```
soukup@host68:~$ ldapmodify -x -W -D "cn=Manager,dc=h68,dc=dat151" -f modify_path.ldif
Enter LDAP Password:
modifying entry "uid=testuser,ou=People,dc=h68,dc=dat151"
```

```
soukup@host68:~$ ldapsearch -x -D "cn=Manager,dc=h68,dc=dat151" -W -b "dc=h68,dc=dat151" "(uid=testuser)"
Enter LDAP Password:
# extended LDIF
#
# LDAPv3
# base <dc=h68,dc=dat151> with scope subtree
# filter: (uid=testuser)
# requesting: ALL
#

# testuser, People, h68.dat151
dn: uid=testuser,ou=People,dc=h68,dc=dat151
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: testuser
sn: User
givenName: Test
cn: Test User
displayName: Test User
uidNumber: 2000
gidNumber: 2000
userPassword:: e1NTSEF9TWJwMmpOVXRuKzlMcENTWjcwUXB6ajk0TGVYSlV2ays=
gecos: Test User
loginShell: /bin/bash
homeDirectory: /share/home/testuser

# search result
search: 2
result: 0 Success

# numResponses: 2
# numEntries: 1
```

## c) Client configuration and verification

First we createt the mount point and then we edited the /etc/fstab to ensure the home directories will still be there after a reboot.

```
ffeilke@host67:~$ sudo mount -t nfs 10.0.0.68:/share/home /share/home
ffeilke@host67:~$ df -h | grep share
10.0.0.68:/share/home                 220G   14G  207G   7% /share/home
ffeilke@host67:~$ sudo vim /etc/fstab
```

```
#
# /etc/fstab
# Created by anaconda on Mon Jan 19 12:28:32 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=4cbf9a96-75ac-4145-b27e-3d5ce2a19268 /                       xfs     defaults        0 0
UUID=838c85fe-19a4-43a9-ae2a-fd5be5e05322 /boot                   xfs     defaults        0 0
UUID=36B9-B457          /boot/efi               vfat    umask=0077,shortname=winnt 0 2
UUID=683bd9b4-47ad-4e26-8ab6-c244ba6e27df none                    swap    defaults        0 0
10.0.0.68:/share/home /share/home nfs defaults 0 0
~
~
```

```
ffeilke@host67:~$ bat /etc/fstab
       File: /etc/fstab

   #
   # /etc/fstab
   # Created by anaconda on Mon Jan 19 12:28:32 2026
   #
   # Accessible filesystems, by reference, are maintained under '/dev/disk/'.
   # See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
   #
   # After editing this file, run 'systemctl daemon-reload' to update systemd
   # units generated from this file.
   #
   UUID=4cbf9a96-75ac-4145-b27e-3d5ce2a19268 /                       xfs     defaults        0 0
   UUID=838c85fe-19a4-43a9-ae2a-fd5be5e05322 /boot                   xfs     defaults        0 0
   UUID=36B9-B457          /boot/efi               vfat    umask=0077,shortname=winnt 0 2
   UUID=683bd9b4-47ad-4e26-8ab6-c244ba6e27df none                    swap    defaults        0 0
   10.0.0.68:/share/home /share/home nfs defaults 0 0
```

We stop blocking the network home folders, clear the sssd cache and restart the ssd service.

```
ffeilke@host67:~$ sudo sss_cache -E
ffeilke@host67:~$ sudo systemctl restart sssd
ffeilke@host67:~$ sudo setsebool -P use_nfs_home_dirs 1
```

For verification we created a testfile on the client and read the same file on the server.

```
ffeilke@host67:~$ su - testuser
Password:
Last login: Tue Apr 14 12:05:21 CEST 2026 on pts/0
testuser@host67:~$ pwd
/share/home/testuser
testuser@host67:~$ echo "Success!" > final_proof.txt
testuser@host67:~$ cat final_proof.txt
Success!
testuser@host67:~$
```

```
-------
soukup@host68:~$ sudo cat /share/home/testuser/final_proof.txt
[sudo] password for soukup:
Success!
```

## Task 3: Custom PAM profile for pam_mount

The authselect tool on AlmaLinux 10 does not provide any ready made feature that enables the PAM module pam_mount. The task here is to create a authselect profile with a a feature that will include pam_mount with SSSD. Observe, the task here is not to use, nor configure pam_mount for logins, but only to create an authselect profile with a feature that adds the pam_mount module to the PAM system. Do not modify PAM files below "/etc/pam.d/" that are managed by the authselect tool. You will need to modify though files below "/etc/authselect/custom/". Be careful not to modify any of the files of "/usr/share/authselect/default/", i.e. the files that you modify must not be soft links. To solve this task you must create a custom profile based on the SSSD profile. Add a feature to the profile that adds the PAM module pam_mount. Name the feature with-mount. The RPM package for the pam_mount module is not available in the default repositories of EL10. I have repacked the Fedora 43 version for EL10. The necessary packages are available from the host eple.hvl.no. The commands below will add the repository on eple to your Alma 10 computer:

### a) Install `pam_mount` package from EPLE repository

On AlmaLinux 10, `pam_mount` is not in the default repositories. We added the EPLE repository and installed the package with `--enablerepo`.

```
cat <<EOF | sudo tee /etc/yum.repos.d/eple.repo >/dev/null
[eple]
name=eple-repo
priority=1
baseurl=http://eple.hvl.no/repos/almalinux-10/x86_64/
skip_if_unavailable=True
enabled=0
EOF

sudo rpm --import http://eple.hvl.no/repos/keys/RPM-GPG-KEY-pmanager
sudo dnf clean all
sudo dnf --enablerepo=eple install pam_mount
```

```
ffeilke@host67:~$ cat <<EOF | sudo tee /etc/yum.repos.d/eple.repo >/dev/null
[eple]
name=eple-repo
priority=1
baseurl=http://eple.hvl.no/repos/almalinux-10/x86_64/
skip_if_unavailable=True
enabled=0
EOF
ffeilke@host67:~$ sudo rpm --import http://eple.hvl.no/repos/keys/RPM-GPG-KEY-pmanager
ffeilke@host67:~$ sudo dnf clean all
33 files removed
ffeilke@host67:~$ sudo dnf --enablerepo=eple install pam_mount
eple-repo                                          175 kB/s |  12 kB     00:00
AlmaLinux 10 - AppStream                           3.7 MB/s | 2.3 MB     00:00
AlmaLinux 10 - BaseOS                               29 MB/s |  19 MB     00:00
AlmaLinux 10 - CRB                                 1.3 MB/s | 533 kB     00:00
AlmaLinux 10 - Extras                               30 kB/s |  12 kB     00:00
Extra Packages for Enterprise Linux 10 - x86_64    3.2 MB/s | 5.7 MB     00:01
Dependencies resolved.
=================================================================================
 Package                Architecture     Version                  Repository         Size
=================================================================================
Installing:
 pam_mount              x86_64           2.20-4.el10              eple              104 k
Installing dependencies:
 hxtools                x86_64           20150304-23.el10         eple               47 k
 libHX                  x86_64           3.22-25.el10             eple               66 k
 libcryptmount          x86_64           2.20-4.el10              eple               21 k
 perl-Filter            x86_64           2:1.64-512.el10          appstream          88 k
 perl-Tie               noarch           4.6-512.2.el10_0         appstream          27 k
 perl-encoding          x86_64           4:3.00-511.el10          appstream          63 k

Transaction Summary
=================================================================================
Install  7 Packages

Total download size: 416 k
Installed size: 933 k
Is this ok [y/N]:
```

The output shows that `pam_mount` and its dependencies were resolved from the EPLE repository.

## b) Create custom authselect profile from SSSD

We created a custom profile named `sssd` with SSSD as base profile:

```
sudo authselect create-profile sssd -b sssd
```

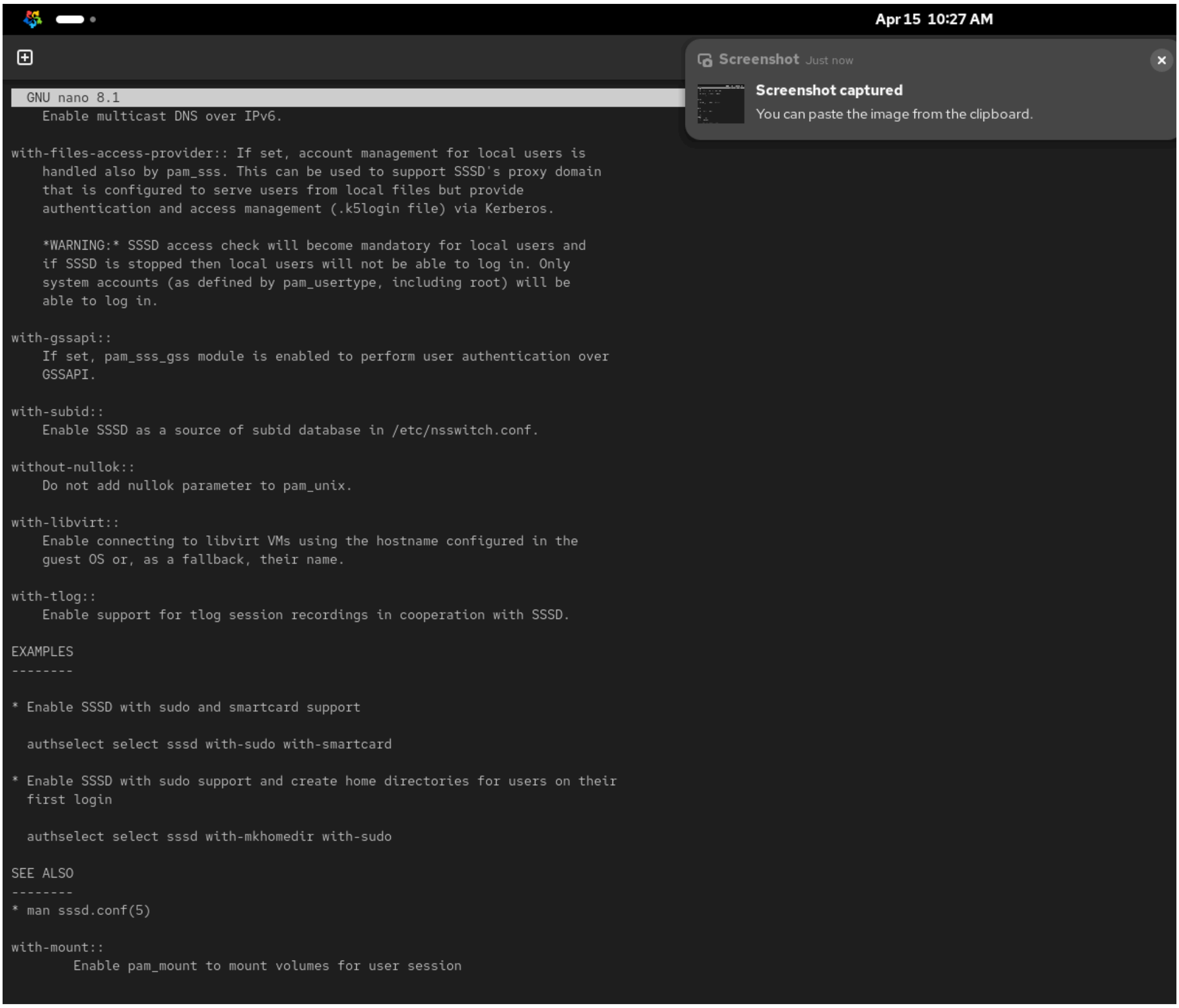
This creates editable files under `/etc/authselect/custom/sssd/` (not symlinks to `/usr/share/authselect/default/`), which is required for safe customization.

```
ffeilke@host67:~$ sudo authselect create-profile sssd -b sssd
New profile was created at /etc/authselect/custom/sssd
ffeilke@host67:~$ nano /etc/authselect/custom/sssd-custom
ffeilke@host67:~$ nano /etc/authselect/custom/sssd/RE
README        REQUIREMENTS
ffeilke@host67:~$ nano /etc/authselect/custom/sssd/RE
README        REQUIREMENTS
ffeilke@host67:~$ nano /etc/authselect/custom/sssd/README
ffeilke@host67:~$ sudo nano /etc/authselect/custom/sssd/README
```

c) Define feature description in profile README

We edited `/etc/authselect/custom/sssd/README` and added the `with-mount` feature description so it appears in the profile feature list.

```
Apr 15 10:26 AM
ffeilke@host67:~ — nano /etc/authselect/custom/sssd/README

 GNU nano 8.1                                   /etc/authselect/custom/sssd/README
Enable SSSD for system authentication (also for local users only)
=================================================================

Selecting this profile will enable SSSD as the source of identity
and authentication providers.

SSSD provides a set of daemons to manage access to remote directories and
authentication mechanisms such as LDAP, Kerberos, FreeIPA or AD. It provides
an NSS and PAM interface toward the system and a pluggable backend system
to connect to multiple different account sources.

More information about SSSD can be found on its project page:
https://sssd.io

However, if you do not want to keep SSSD running on your machine, you can
keep this profile selected and just disable SSSD service. The resulting
configuration will still work correctly even with SSSD disabled and local users
and groups will be read from local files directly.

SSSD CONFIGURATION
------------------

Authselect does not touch SSSD's configuration. Please, read SSSD's
documentation to see how to configure it manually. Only local users
will be available on the system if there is no existing SSSD configuration.

AVAILABLE OPTIONAL FEATURES
---------------------------

with-faillock::
    Enable account locking in case of too many consecutive
    authentication failures.

with-mkhomedir::
    Enable automatic creation of home directories for users on their
    first login.

with-smartcard::
    Enable authentication with smartcards through SSSD. Please note that
    smartcard support must be also explicitly enabled within
    SSSD's configuration.

with-smartcard-lock-on-removal::
    Lock screen when a smartcard is removed.

with-smartcard-required::
    Smartcard authentication is required. No other means of authentication
    (including password) will be enabled.
```




```
GNU nano 8.1
    Enable multicast DNS over IPv6.

with-files-access-provider:: If set, account management for local users is
    handled also by pam_sss. This can be used to support SSSD's proxy domain
    that is configured to serve users from local files but provide
    authentication and access management (.k5login file) via Kerberos.

    *WARNING:* SSSD access check will become mandatory for local users and
    if SSSD is stopped then local users will not be able to log in. Only
    system accounts (as defined by pam_usertype, including root) will be
    able to log in.

with-gssapi::
    If set, pam_sss_gss module is enabled to perform user authentication over
    GSSAPI.

with-subid::
    Enable SSSD as a source of subid database in /etc/nsswitch.conf.

without-nullok::
    Do not add nullok parameter to pam_unix.

with-libvirt::
    Enable connecting to libvirt VMs using the hostname configured in the
    guest OS or, as a fallback, their name.

with-tlog::
    Enable support for tlog session recordings in cooperation with SSSD.

EXAMPLES
--------

* Enable SSSD with sudo and smartcard support

  authselect select sssd with-sudo with-smartcard

* Enable SSSD with sudo support and create home directories for users on their
  first login

  authselect select sssd with-mkhomedir with-sudo

SEE ALSO
--------
* man sssd.conf(5)

with-mount::
        Enable pam_mount to mount volumes for user session
```

The feature is documented as:

```
with-mount::
    Enable pam_mount to mount volumes for user session
```

This is important because the assignment requires the feature to be listed and documented by `authselect`.

### d) Add `pam_mount` through feature guards in PAM templates

We edited `/etc/authselect/custom/sssd/system-auth` and added `pam_mount.so` in the required places guarded by the feature flag:

- In `auth` section (before `pam_deny.so`): `auth optional pam_mount.so {include if "with-mount"}`
- In `session` section: `session optional pam_mount.so {include if "with-mount"}`

```
GNU nano 8.1                                           /etc/authselect/custom/sssd/system-auth
auth        [success=done ignore=ignore default=die]     pam_sss.so require_cert_auth ignore_authinfo_unavail   {include if "with-smartcard-required"}
auth        sufficient                                   pam_fprintd.so                                         {include if "with-fingerprint"}
auth        sufficient                                   pam_u2f.so cue                                         {include if "with-pam-u2f"}
auth        required                                     pam_u2f.so cue {if not "without-pam-u2f-nouserok":nouserok} {include if "with-pam-u2f-2fa"}
auth        [default=1 ignore=ignore success=ok]         pam_usertype.so isregular
auth        [default=1 ignore=ignore success=ok]         pam_localuser.so                                       {exclude if "with-smartcard"}
auth        [default=2 ignore=ignore success=ok]         pam_localuser.so                                       {include if "with-smartcard"}
auth        [success=done authinfo_unavail=ignore user_unknown=ignore ignore=ignore default=die] pam_sss.so try_cert_auth {include if "with-smartcard"}
auth        sufficient                                   pam_unix.so {if not "without-nullok":nullok}
auth        [default=1 ignore=ignore success=ok]         pam_usertype.so isregular                              {include if "with-gssapi"}
auth        sufficient                                   pam_sss_gss.so                                         {include if "with-gssapi"}
auth        [default=1 ignore=ignore success=ok]         pam_usertype.so isregular
auth        sufficient                                   pam_sss.so forward_pass
auth        required                                     pam_faillock.so authfail                               {include if "with-faillock"}

auth        optional                                     pam_mount.so                                           {include if "with-mount"}

auth        optional                                     pam_gnome_keyring.so only_if=login auto_start          {include if "with-pam-gnome-keyring"}
auth        required                                     pam_deny.so

account     required                                     pam_access.so                                          {include if "with-pamaccess"}
account     required                                     pam_faillock.so                                        {include if "with-faillock"}
account     required                                     pam_unix.so
account     sufficient                                   pam_localuser.so                                       {exclude if "with-files-access-provider"}
account     sufficient                                   pam_usertype.so issystem
account     [default=bad success=ok user_unknown=ignore] pam_sss.so
account     required                                     pam_permit.so

password    requisite                                    pam_pwquality.so local_users_only
password    [default=1 ignore=ignore success=ok]         pam_localuser.so                                       {include if "with-pwhistory"}
password    requisite                                    pam_pwhistory.so use_authtok                           {include if "with-pwhistory"}
password    sufficient                                   pam_unix.so yescrypt shadow {if not "without-nullok":nullok} use_authtok
password    [success=1 default=ignore]                   pam_localuser.so
password    sufficient                                   pam_sss.so use_authtok
password    required                                     pam_deny.so

session     optional                                     pam_keyinit.so revoke
session     required                                     pam_limits.so
-session    optional                                     pam_systemd.so
session     optional                                     pam_oddjob_mkhomedir.so                                {include if "with-mkhomedir"}
session     [success=1 default=ignore]                   pam_succeed_if.so service in crond quiet use_uid
session     required                                     pam_unix.so
session     optional                                     pam_sss.so

session     optional                                     pam_mount.so                                           {include if "with-mount"}

session     optional                                     pam_gnome_keyring.so only_if=login auto_start          {include if "with-pam-gnome-keyring"}
```

This ensures `pam_mount` is injected only when the `with-mount` feature is enabled.

## e) Select custom profile and enable feature

After editing the profile, we selected the custom profile with the `with-mount` feature and verified it:

```
authselect list
sudo authselect select custom/sssd with-mount --force
authselect current
```

```
ffeilke@host67:~$ sudo nano /etc/authselect/custom/sssd/system-auth
ffeilke@host67:~$ authselect list
- local          Local users only
- sssd           Enable SSSD for system authentication (also for local users only)
- winbind        Enable winbind for system authentication
- custom/sssd    Enable SSSD for system authentication (also for local users only)
ffeilke@host67:~$ sudo authselect select custom/sssd with-mount --force
[sudo] password for ffeilke:
Backup stored at /var/lib/authselect/backups/2026-04-15-08-37-40.GFcmqZ
Profile "custom/sssd" was selected.

Make sure that SSSD service is configured and enabled. See SSSD documentation for more information.

ffeilke@host67:~$ authselect current
Profile ID: custom/sssd
Enabled features:
- with-mount
ffeilke@host67:~$
```

The output confirms:

- `custom/sssd` appears in `authselect list`
- Profile ID is `custom/sssd`
- `with-mount` is listed under enabled features

This proves the custom profile and feature were created correctly and activated.

---

## Task 4: Decoding PAM rules

In task three, PAM rules for a SSSD profile that includes the pam_mount module was shown. Explain the consequences of line four in the listing of PAM rules, the yellow line with pam_usertype.so. What would have been the consequences if the line was replaced with the line below?

### Rule under analysis

```
auth [default=1 ignore=ignore success=ok] pam_usertype.so isregular
```

```
auth     required                                  pam_env.so
auth     required                                  pam_faildelay.so delay=2000000
auth     required                                  pam_faillock.so preauth silent
auth     [default=1 ignore=ignore success=ok]      pam_usertype.so isregular
auth     [default=1 ignore=ignore success=ok]      pam_localuser.so
auth     sufficient                                pam_unix.so nullok
auth     [default=1 ignore=ignore success=ok]      pam_usertype.so isregular
auth     sufficient                                pam_sss.so forward_pass
auth     required                                  pam_faillock.so authfail
auth     optional                                  pam_mount.so
auth     required                                  pam_deny.so
```

### Consequences of the original rule

`pam_usertype.so isregular` checks whether the account is a regular user.

Control behavior:

- `success=ok`: if user is regular, authentication continues to the next line normally.
- `default=1`: if the result is neither success nor ignore, PAM skips one following rule.
- `ignore=ignore`: ignore leaves flow unaffected.

In this instance, the effect is following:

- Regular user: continue with `pam_localuser.so`, then continue through the stack.
- Non-regular/system-type user: skip one entry (the `pam_localuser.so` line), then continue at the next rule (`pam_unix.so`).

So the line is used to slightly alter control flow for user categories without immediately denying authentication.

### If replaced by `auth [default=ok success=6] pam_usertype.so isregular`

```
auth [default=ok success=6] pam_usertype.so isregular
```

Consequences:

- If `isregular` succeeds, PAM skips 6 following `auth` lines.
- In the shown order, that jump lands at the final deny path (`pam_deny.so`), causing regular users to fail authentication.
- If `isregular` does not succeed, `default=ok` means no skip and flow continues normally for the non-regular case.

This replacement effectively inverts intended behavior and would block normal regular-user logins.